ABHISHEK RAMAMURTHY RATTIHALLI

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Summary

MCA graduate with a strong foundation in software development, data science, machine learning and backend application development. Skilled in Python, Java, SQL, Spring Boot, FastAPI, Pandas, NumPy and Scikit-learn, with hands-on experience in exploratory data analysis, feature engineering, predictive modelling, REST API development and software design. Interested in building scalable, intelligent and user-focused applications using machine learning and modern backend technologies.

Education

R V Institute of Technology — Bengaluru, Karnataka
Master of Computer Applications (MCA) — 2024 – 2026

Government First Grade College — Hirekerur, Karnataka
Bachelor of Science (B.Sc.) — 2020 – 2024

Technical Skills

Programming: Python, Java, SQL, C, Object-Oriented Programming
Machine Learning & Data Science: Scikit-learn, Pandas, NumPy, Exploratory Data Analysis, Feature Engineering, Model Training, Model Evaluation, Predictive Modelling, NLP
Backend & Development: FastAPI, Spring Boot, Hibernate, REST APIs, Microservices
Databases: MySQL, PostgreSQL, MongoDB
DevOps & Tools: Git, GitHub, Docker, Jupyter, Power BI, Microsoft Excel, MLflow, Snowflake
Cloud: AWS

Projects

Real-Time Customer Churn & Lifetime Value (LTV) Engine

- Built an end-to-end customer risk analytics platform to predict subscriber churn and estimate customer financial risk for SaaS and e-commerce use cases.
- Performed RFM (Recency, Frequency, Monetary) analysis and feature engineering using Python, Pandas and NumPy, followed by churn modelling using XGBoost and LightGBM.
- Developed an interactive Streamlit dashboard for customer segmentation, risk scoring, churn probability and LTV insights.

Automated Document Intelligence & RAG Pipeline

- Developed a document intelligence pipeline to extract, retrieve and summarize information from unstructured enterprise PDFs and financial reports.
- Implemented document chunking, embeddings, semantic search and Retrieval Augmented Generation (RAG) using LangChain and vector databases such as Chroma/Pinecone.
- Built FastAPI services for document ingestion, retrieval and question answering, with modular Python components for scalable deployment.

Stock Trend Prediction System

- Developed a Python-based stock trend prediction application using historical market data, technical indicators and machine learning techniques to identify potential upward and downward price trends.
- Performed data preprocessing, exploratory analysis, feature engineering and model evaluation using Pandas, NumPy and Scikit-learn.
- Built an interactive Streamlit dashboard to visualize historical prices, technical indicators, model predictions and trend signals.

AI Chatbot with Streamlit

- Developed an AI-powered conversational chatbot using Python and Streamlit with an interactive web interface for real-time user conversations.
- Implemented prompt handling, conversational context and response generation through an LLM-based workflow.

- Designed the application with a modular Python backend and Streamlit frontend, focusing on clean interaction flow and maintainable application structure.

Certifications

- AWS Certified Cloud Practitioner – Amazon Web Services
- Introduction to Applied Data Science with Python – Simplilearn
- Python for AI Projects: From Data Exploration to Impact – LinkedIn Learning
- Python for Data Science and Machine Learning Essentials – LinkedIn Learning
- Data Engineering Professional Certificate – Snowflake and LinkedIn Learning

Extracurricular Activities

- Core Organizer for the MCA "Advaya" Tech Fest, serving as Marketing Head.
- Volunteer Educator with U&I NGO, supporting education for underprivileged youth.

Links

GitHub: github.com/D0ACE